Concept Check: Introduction to Transfer Learning

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1. Which of the following is a significant advantage of transfer learning? **1 / 1 point**

- ○ Utilizes models developed by experts
- ○ Initially trained on extensive image datasets
- ○ Faster training speeds than starting from scratch
- ◉ All of the above

Correct
Transfer learning saves time and resources by leveraging preexisting models, with architecture developed by experts and weights determined often from millions of images.

2. True or False: A model initially trained on a large, general-purpose dataset **should not** be used in transfer learning for a specific application like classifying traffic signs. **1 / 1 point**

- ◉ False
- ○ True

Correct
Deep learning models trained on large datasets of general image classes use general image features like edges, textures, and blobs, which apply to most image classification tasks.

3. How does transfer learning enable faster and more accurate training than training from scratch? **1 / 1 point**

- ◉ By using preexisting weights as a starting point
- ○ By increasing the number of layers in the model
- ○ By re-designing the entire model architecture

Correct
Transfer learning starts with preexisting weights, enabling faster and more accurate training by building knowledge established by training on thousands or millions of other images.

4. Which is **not** usually a factor in choosing which pre-trained network to use as a starting point? **1 / 1 point**

- ◉ The number of neurons in the fully-connected layer needs to match the number of classes in your dataset
- ○ Model prediction speed
- ○ Accuracy on a benchmark dataset

Correct
You will replace the image classification layer to match your dataset.